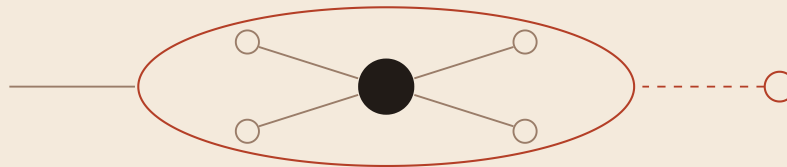


A High-Level View of the Schema of Inquiry



1. Inquiry changes what an object can be

Scientific work does not only accumulate facts about objects that are already fully known. It also changes the access through which an object can be recognized. A new instrument, experiment, language, or comparison may separate candidates that once looked identical. A new explanation may move in the other direction, showing that several presentations belong together for a reason.

The repository calls a historically available way of observing, interacting with, or translating an object a **lens**. A family of lenses produces a current observational profile. That profile can support comparison, but it cannot quietly become a final account of the object. Its reach is declared, and the possibility of another lens remains visible.

This turns the protagonist of inquiry into something revisable: not an object with all its properties laid out in advance, and not an object reduced to measurements, but an object provisionally stabilized relative to present access and a present task. A claim may be sufficient for identification or reconstruction under those conditions without claiming that no future distinction is possible.

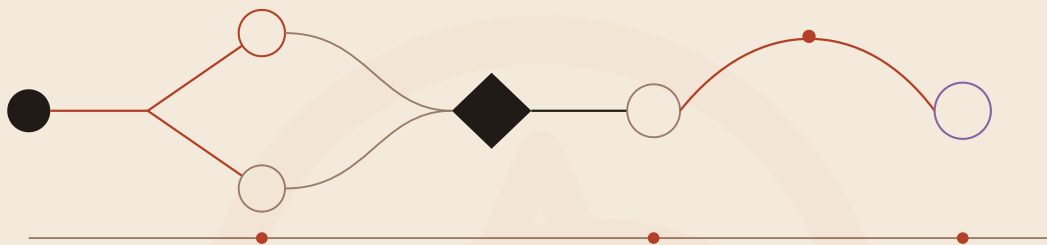
The governing question is therefore not whether one formalism can name all of science. It is whether the movements by which inquiry separates, identifies, connects, and reorganizes objects can be made precise enough to test without being frozen into a terminal picture.

EVIDENCE BOUNDARY. This is an exploratory research program. Ordinary Yoneda is a disciplined comparison point for relational completeness inside a declared category; it is not evidence that scientific lenses already form the required category or that science as a whole is Yoneda.

2. Separate, identify, transport

Once a lens family and task are declared, inquiry can move in three directions that should not be confused. A lens may **separate** candidates by revealing a difference. A justified identification may **unify** presentations, but only with an explicit license such as an equivalence, invariant, universal property, or transport law. A bridge may **transport** a result, constraint, or obstruction from one domain into another.

These are not merely three degrees of the same kind of progress. Separation changes discriminatory reach. Identification changes explanatory organization. Transport changes which consequences are reachable across domains. Each movement has its own success condition, and each can fail while the others succeed.



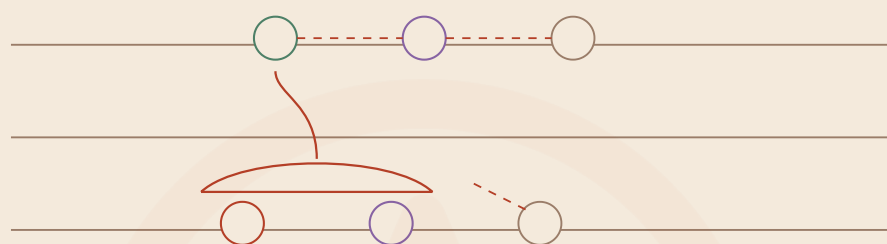
Residue is load-bearing. It records what an identification did not preserve, what a translation did not carry, or what a lens could not distinguish. Without it, an elegant unification can conceal a lossy quotient and a bridge can collapse into a change of vocabulary.

The project therefore refuses a single total information score. More distinctions, fewer presentations, and farther-reaching consequences may each matter, but no common ordering has been earned. The gain lies first in making the new distinction, the licensing reason, the transported consequence, and the unpreserved remainder accountable.

3. The schema is a revisable achievement

Once identifications and bridges are available, inquiry can ask a different question: not only *what is this?*, but *of what is this an instance?* A schema is more than a label for similarities. It must admit some candidates, exclude others, and support at least one non-trivial transfer, prediction, or constraint among the things it organizes.

An instance is not merely placed under a heading. It comes with an interpretation showing how its relevant structure realizes the schema. That interpretation can later fail, expose residue, or be revised. The schema earns its place by doing work while remaining answerable to the cases from which it was abstracted.



DISCRIMINATORY INFORMATION

Which candidates can the declared lenses distinguish?

STRUCTURAL INFORMATION

What licensed reason lets several presentations count as one pattern?

TRANSPORT INFORMATION

Which claims, constraints, or obstructions cross a bridge, and under what conditions?

The recursive picture matters because abstraction changes the field of possible inquiry. A schema that organizes one level can become an object to be compared at the next. New bridges may then appear between schemas, and their transported constraints may expose parts of the lower-level cases that had not yet been visible.

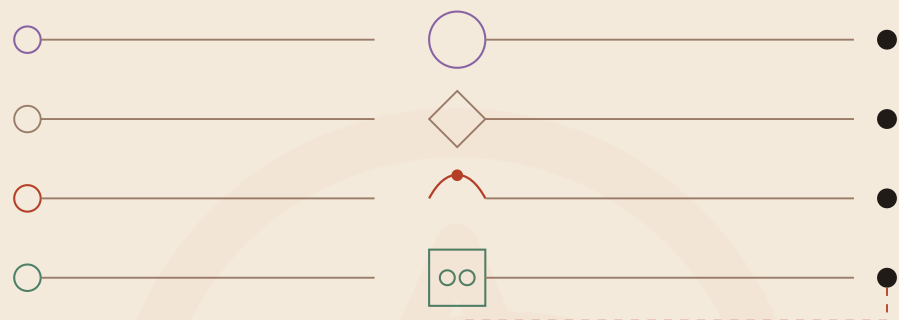
But recurrence is not convergence. Nothing in the repository establishes a final universal taxonomy, a complete family of scientific lenses, or a single information order. The intended achievement is smaller and harder: a progressive account in which every abstraction records how it was earned and remains vulnerable to the next discriminating lens.



4. What would make it more than a metaphor

The project is designed to resist attractive but empty generality. A proposed lens, identification, bridge, or schema is admitted only when it comes with the information needed to test what it claims. The contracts below are not final mathematics. They are minimum evidence requirements that keep the vocabulary from carrying the argument by itself.

A bounded inquiry run begins by naming candidates, lenses, a task, observations, and known limits. From there, each stronger movement crosses its own gate. Failure codes reject unsupported claims rather than declaring that no better construction could exist.



CURRENT STATUS

No central thesis is proved and no Lean witness exists yet. The repository contains governed vocabulary, a layered system view, candidate contracts, collapse tests, essays, and visual explanations. Ordinary Yoneda remains owned precedent reported outside this repository; its extension to scientific inquiry is open.

RELATIVE, NOT FINAL

Adequacy is always parameterized by declared access and a declared task.

POWER WITH RESIDUE

Every distinction, unification, and transport states what it reveals and what it leaves behind.

RECURSIVE, NOT TERMINAL

A schema may become an instance at another level without implying one final schema of everything.

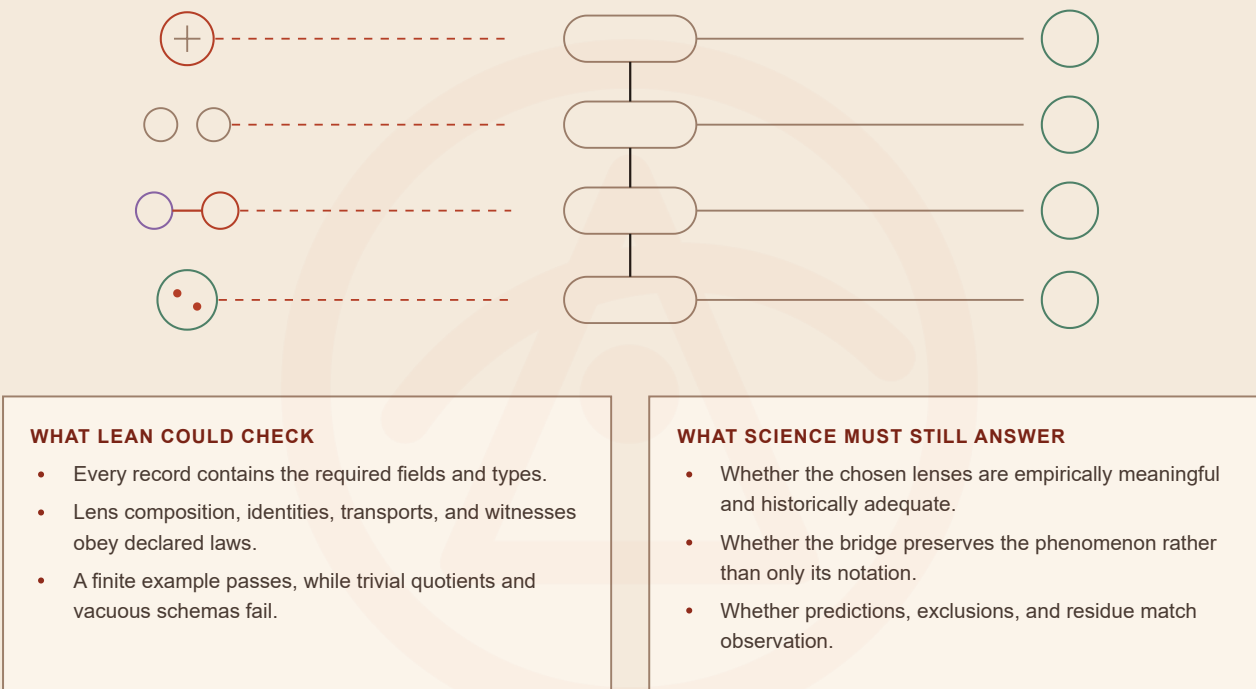
Based on `schema-of-inquiry/README.md` and `views/ontology-view.md`, `system-view.md`, and `engineer-view.md`.



5. A first formal shape, in plain language

Formalization means replacing suggestive words with objects whose parts and obligations are explicit. It does not mean that scientific inquiry has already been reduced to mathematics. The first target is deliberately modest: define a small grammar for one inquiry run, then test whether concrete examples can inhabit it without making the framework trivial.

An **InquiryRun** would record the candidates under comparison, the available lens family, the task, the observations, and known limits. Stronger moves would require stronger records. An **IdentificationWitness** would carry the reason for unification and its residue. A **BridgeWitness** would carry a claim across domains under a preservation condition. A **SchemaCandidate** would state what it admits, what it excludes, what it enables, and what would force revision.



Formal verification can establish that a proposed construction is internally well-formed and that a stated consequence follows from stated assumptions. It cannot, by itself, decide whether the assumptions describe nature, whether the lenses matter to a scientific community, or whether the resulting explanation is worth keeping. The formal and empirical layers constrain one another without collapsing into one another.

Formalization status: planned Layer 1. No local Lean definitions or checked examples exist yet.